Calculus II Lesson 01

Question 1:

Find $\int 4(x-15)^6 dx$ Hint: Let $u = x - 15$

a) $du =$ ______________________________

b) Answer: $\int 4(x-15)^6 dx =$ ______________________________

Question 2:

Find $\int \frac{17}{(x-1)^7} dx$ Hint: Let $u = x - 1$

a) $du =$ ______________________________

b) Answer: $\int \frac{17}{(x-1)^7} dx =$ ______________________________

Question 3:

Find $\int \left[x + \frac{1}{(2x-11)^3} \right] dx = \int x dx + \int \frac{1}{(2x-11)^3} dx =$

$a) \int \left[x + \frac{1}{(2x-11)^3} \right] dx =$ ______________________________

$b) \int_0^2 \left[x + \frac{1}{(2x-11)^3} \right] dx =$ ______________________________

Question 4:

Find $\int \frac{x^2-5}{-x^3+15x+1} dx = \int \frac{1}{-x^3+15x+1} \left(x^2-5 \right) dx$

Hint: Let $u = -x^3 + 15x + 1$ $\quad \frac{du}{dt} = -3x^2 + 15 = -3(x^2-5)$

a) $du =$ ______________________________

b) $\int \frac{x^2-5}{-x^3+15x+1} dx =$ ______________________________

Question 5:

Find $\int \frac{4x^2}{x-1}dx$ Hint: Divide $\frac{4x^2}{x-1}$ first.

a) Divide $\frac{4x^2}{x-1} = ?$

b) Answer: $\int \frac{4x^2}{x-1}dx =$ ______________________________

Question 6:

Find $\int \frac{e^x}{10+e^x}dx = \int \frac{1}{10+e^x}\left(e^x\right)dx$ Hint: Let $u = 10 + e^x$

a) $du =$ ______________________________

b) Answer: $\int \frac{e^x}{10+e^x}dx =$ ______________________________

Question 7:

Find $\int x \cdot \sin\left(4\pi x^2\right) dx$ Hint: Let $u = 4\pi x^2$

a) $du =$ ______________

b) $\int x \cdot \sin\left(4\pi x^2\right) dx =$ ____________________________

Question 8:

Find $\int \frac{\cos x}{\sqrt{\sin x}} dx = \int \frac{1}{\sqrt{\sin x}} (\cos x) dx$ Hint: Let $u = \sin x$

a) $du =$ ____________________________

b) $\int \frac{\cos x}{\sqrt{\sin x}} dx =$ ____________________________

Question 9:

Find $\int \frac{\ln x^2}{x} dx = \int \ln x^2 \cdot \left(\frac{1}{x}\right) dx$ Hint: Let $u = \ln x^2 = 2\ln x$

a) $du =$ ______________________________

b) $\int \frac{\ln x^2}{x} dx =$ ______________________________

Question 10:

$\int \frac{8x}{\sqrt{x^2+16}} dx = \int \frac{1}{\sqrt{x^2+16}} (8x) dx$ Hint: Let $u = x^2 + 16$

a) $du =$ ______________________________

b) $\int \frac{8x}{\sqrt{x^2+16}} dx =$ ______________________________

Question 11:

$$\int \frac{1}{13+4x^2}dx \qquad \text{Hint: Use Formula } \int \frac{1}{a^2+u^2}du = \frac{1}{a}\arctan\frac{u}{a} + C$$

a) $a^2 =$ ______________________________

b) $u^2 =$ ______________________________

c) $\int \frac{1}{13+4x^2}dx =$ ______________________________

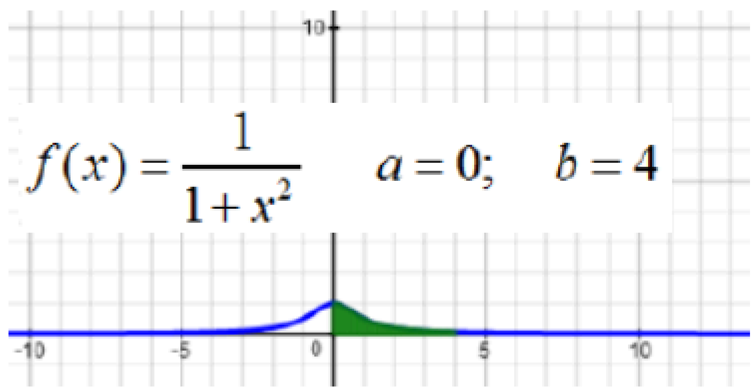
Question 12:

12) Find the area. Hint: Use Formula $\int \frac{1}{a^2+u^2}du = \frac{1}{a}\arctan\frac{u}{a} + C$

$$f(x) = \frac{1}{1+x^2} \qquad a = 0; \quad b = 4$$

a) Set up definite integral: Area $= \int_{?}^{?} \; ? \; dx$

b) Evaluate integral: Area = ____________________



Question 13:

13) Find the area.

$$f(x) = \frac{1}{\sqrt{x^2 + 0.4}} \qquad a = 0; \quad b = 2$$

a) Set up definite integral: Area $= \int_{?}^{?} \; ? \; dx$

b) Evaluate integral: Area = ________________